

Does financial education improve loan performance?

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Background

- Two components of consumer protection reform since 2008:
 - Mandating disclosures by financial firms
 - Improving financial literacy
- Evidence on effectiveness of financial literacy remains inconclusive.

The question

- Households seen to be making unwise decisions, especially w.r.t credit.
- Low financial literacy seen to be a big contributor to bad decisions, and sub-optimal outcomes
- Question: Does financial literacy improve loan performance of micro-finance clients?

Why is this important?

- Access to finance is possible when people are able to access information and act on the same
- Empowering consumers can improve access

Outline of the talk

- Setting
 - Literature
 - The firm and the financial education initiative
- Data
- Results

Part I

The setting

Literature: Effect on repayment and default

- *Gerardi, Goette, Meier 2010*: Numerical ability negatively correlated with delinquency and default; Financial illiteracy played an important role in the sub-prime mortgage crisis
- *Agarwal et. al. 2010*: Participants in a voluntary financial education program less likely to fall behind on their mortgage payments
- *Paxton, Graham & Cameron 2000*: In Burkina Faso, micro-finance groups that received loan literacy training had higher repayment performance

Literature: Group homogeneity

- *Theoretically*: Social connections in group lending work for a few reasons (Karlan, 2007)
 - Individuals are able to select creditworthy peers
 - are able to monitor each other's use of funds
 - are able to enforce repayment
 - are more likely to repay merely because of altruism towards those in their group
- *Empirically*: ambiguous effects of group homogeneity on financial outcomes

The financial firm: Swadhaar

- NBFC providing micro-finance
- Headquartered in Mumbai, branches in Mumbai and cities of Gujarat
- Provides financial services to economically active urban poor

The financial education initiative

- Client Education Program (CEP) is a mandatory one and a half hour programme
- The session is conducted in the branch itself prior to the disbursement of the loan
- Trainers are graduates or undergraduates and mostly women
- The key objective of this initiative is that clients improve their ability to manage credit and become financially informed about savings and rights as customers.
- The program also educates the client to move beyond credit and manage their finances well for better financial planning for future

Sequencing of financial literacy

Branch name	City/State	Date
Gotri	Gujarat	12 July 2012
Anand	Gujarat	18 July 2012
Thane	Mumbai	27 August 2012
Bharuch	Gujarat	
Wadala	Mumbai	

Part II

Data

Data

- Administrative data from Swadhaar
- Information on repayment of the first six installments
- Socio-economic characteristics of each customer

Branch wise summary statistics

Branch	N	Financial Training	Age	Married	Caste: SC/ST/OBC	Income	No. children	No. dependents	Illiterate
Anand	4516	0.96 (0.20)	37.39 (8.40)	0.92 (0.27)	0.17 (0.38)	7997.25 (1440.12)	1.37 (1.22)	2.65 (1.40)	0.49 (0.50)
Bharuch	4008	0.00 (0.00)	36.94 (8.80)	0.83 (0.37)	0.38 (0.49)	7222.08 (1536.94)	1.21 (1.20)	2.20 (1.23)	0.33 (0.47)
Gotri	4035	0.94 (0.25)	37.53 (8.24)	0.90 (0.30)	0.73 (0.44)	9270.94 (901.25)	1.25 (1.14)	2.29 (1.19)	0.37 (0.48)
Thane	3190	0.88 (0.33)	38.41 (8.28)	0.84 (0.37)	0.33 (0.47)	9206.66 (920.53)	1.15 (1.11)	2.27 (1.27)	0.34 (0.47)
Wadala	2147	0.00 (0.00)	38.67 (8.38)	0.76 (0.43)	0.43 (0.49)	9348.12 (740.13)	1.25 (1.16)	2.47 (1.31)	0.26 (0.44)
All	17896	0.61 (0.49)	37.65 (8.45)	0.86 (0.34)	0.40 (0.49)	8470.3 (1487.8)	1.25 (1.17)	2.37 (1.29)	0.372 (0.48)

Approach of measurement

- Use the timing of the financial literacy program across the various branches to identify if there was a causal improvement in the loan repayment performance
- Assumption: the change in average loan repayment performance over time would have been the same at all branches in the absence of the education program
- Any difference between the loan performance in branches with and without financial literacy is attributed to the literacy program.

Measures of performance

- The number of days late for each installment due
- The total number of days late over six installments
- The total number of months delayed repayment

Summary statistics: Loan repayment

	Full sample	Literacy	
		Yes	No
N	17896	10901	6995
Days late installment #1	-0.06 (3.07)	-0.21 (2.13)	0.17 (4.13)
Days late installment #2	-0.04 (3.36)	-0.09 (2.98)	0.03 (3.88)
Days late installment #3	-0.08 (5.09)	-0.10 (4.92)	-0.05 (5.34)
Days late installment #4	-0.03 (6.27)	-0.02 (5.89)	-0.03 (6.82)
Days late installment #5	-0.28 (6.43)	-0.32 (5.33)	-0.21 (7.85)
Days late installment #6	-0.29 (8.23)	-0.37 (7.51)	-0.18 (9.24)
Total Days Late	-0.78 (19.69)	-1.10 (16.69)	-0.27 (23.61)
Number of months late	0.11 (0.40)	0.06 (0.26)	0.18 (0.54)

Part III

Results

Regression results: Each installment

	(1) First Installment	(2) Second Installment	(3) Third Installment	(4) Fourth Installment	(5) Fifth Installment	(6) Sixth Installment
Financial Literacy	-0.256*	-0.303*	-0.475**	-0.617**	-0.792***	-1.101***
	(0.143)	(0.163)	(0.235)	(0.301)	(0.302)	(0.371)
Disbursed Amount	-0.000	-0.000	-0.000**	-0.000***	-0.000	-0.000*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Loan Series	-0.164***	-0.098**	0.061	0.220**	0.114	0.169
	(0.042)	(0.048)	(0.069)	(0.089)	(0.089)	(0.109)
Married	-0.058	0.045	0.211*	0.101	0.067	0.291*
	(0.068)	(0.078)	(0.112)	(0.143)	(0.144)	(0.177)
Age of Client	0.040*	-0.035	0.005	0.071	0.017	0.099*
	(0.022)	(0.025)	(0.037)	(0.047)	(0.047)	(0.058)
Age Squared	-0.001*	0.000	-0.000	-0.001	-0.000	-0.001
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)
No Education	0.046	0.020	0.054	0.213**	0.052	0.098
	(0.049)	(0.056)	(0.081)	(0.103)	(0.104)	(0.128)
Monthly Income	0.000**	0.000***	0.000	0.000	-0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Number of Earning Members	-0.051	-0.077	-0.036	0.003	-0.066	-0.135
	(0.043)	(0.049)	(0.070)	(0.090)	(0.090)	(0.111)
Reserved Caste	0.112*	-0.134**	0.087	0.154	0.046	0.062
	(0.050)	(0.057)	(0.082)	(0.105)	(0.106)	(0.130)
Number of Dependents	-0.010	-0.016	-0.061**	-0.045	0.032	0.018
	(0.018)	(0.020)	(0.029)	(0.037)	(0.038)	(0.046)
Branch Dummies	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16272	16272	16272	16272	16272	16272
R-Squared	0.03	0.03	0.02	0.04	0.02	0.01

* Significant at the 10%; ** at the 5%; *** at 1% levels

Regressions control for the duration of the loan i.e. 12 months or 24 months

All regressions control for year, month and state fixed effects

Interpreting these results

- Decrease in # of days repayment was delayed in each of the six installments
- The magnitude of the effect increases with each installment
- Financially literate individual are repaying the first installment 0.25 days earlier and the sixth installment one day earlier

Regression results: Over six months

	(1) Total Days Late	(2) Total Days Late	(3) Number of Months Late	(4) Number of Months Late
Financial Literacy	-1.999*** (0.340)	-3.544*** (0.910)	-0.141*** (0.007)	0.052*** (0.020)
Disbursed Amount	-0.000* (0.000)	-0.000*** (0.000)	0.000 (0.000)	-0.000 (0.000)
Loan Series	0.187 (0.268)	0.301 (0.268)	-0.001 (0.006)	0.001 (0.006)
Married	0.457 (0.433)	0.657 (0.433)	-0.019** (0.009)	-0.008 (0.009)
Age of Client	0.208 (0.142)	0.197 (0.142)	0.007** (0.003)	0.006** (0.003)
Age Squared	-0.003 (0.002)	-0.003 (0.002)	-0.000** (0.000)	-0.000* (0.000)
No Education	0.498 (0.313)	0.482 (0.313)	-0.013** (0.007)	-0.010 (0.007)
Monthly Income	0.001*** (0.000)	0.000 (0.000)	0.000*** (0.000)	-0.000 (0.000)
Number of Earning Members	-0.439 (0.273)	-0.362 (0.272)	-0.015** (0.006)	-0.014** (0.006)
Reserved Caste	0.884*** (0.299)	0.327 (0.319)	0.034*** (0.006)	0.005 (0.007)
Number of Dependents	-0.051 (0.113)	-0.082 (0.113)	0.002 (0.002)	0.002 (0.002)
Branch Dummies	No	Yes	No	Yes
Observations	16272	16272	16272	16272
R-Squared	0.02	0.03	0.03	0.05

* Significant at the 10%; ** at the 5%; *** at 1% levels

Regressions control for the duration of the loan i.e. 12 months or 24 months

All regressions control for year, month and state fixed effects

Interpreting these results

- Total number of days repayment is delayed decreases by approximately 2 to 3.5 days
- Months variable is positive: two explanations
 - It is cheaper to spread the interest burden over more months than to pay a high interest every month
 - Positive coefficient is in fact being driven by a particular branch since it switches signs with branch fixed effects

Group homogeneity

	(1) Total Days Late	(2) Number of Months Late
Financial Literacy	-3.117*** (1.002)	0.070*** (0.022)
All Reserved*Financial Literacy	0.994 (0.808)	-0.033* (0.017)
All Reserved Caste	-0.887 (0.627)	0.029** (0.014)
All General*Financial Literacy	-1.409** (0.695)	-0.022 (0.015)
All General Caste	-0.591 (0.510)	-0.002 (0.011)
Observations	16264	16264

* Significant at the 10%; ** at the 5%; *** at 1% levels

Regressions control for the duration of the loan i.e. 12 months or 24 months and demographic variables

All regressions control for year, month, branch and state fixed effects

Interpreting these results

- Low levels of general literacy at about 42% for SC and 34% for ST
- Low literacy levels lead to limited financial awareness and thus adverse financial outcomes
- If all MFI group members belong to the same backward caste, there is a detrimental effect on repayment time.
- This effect goes away after financial literacy classes

Threats to validity

- Drop one branch at a time
- Propensity score matching to match on observables
- The results do not change

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Part IV

Conclusions

Conclusion

- Results suggest that financial literacy led to a decline in the total number of days taken to make loan repayments as well as the number of months in which repayment is delayed
- Seems to have helped those in homogeneous groups of reserved castes overcome their initial disadvantage of low financial literacy relative to those in mixed caste groups.

Channel of change

- Direct vs. indirect
- Even if financial literacy only improves financial product awareness and individual attitudes towards making financial decisions, it is an important step